

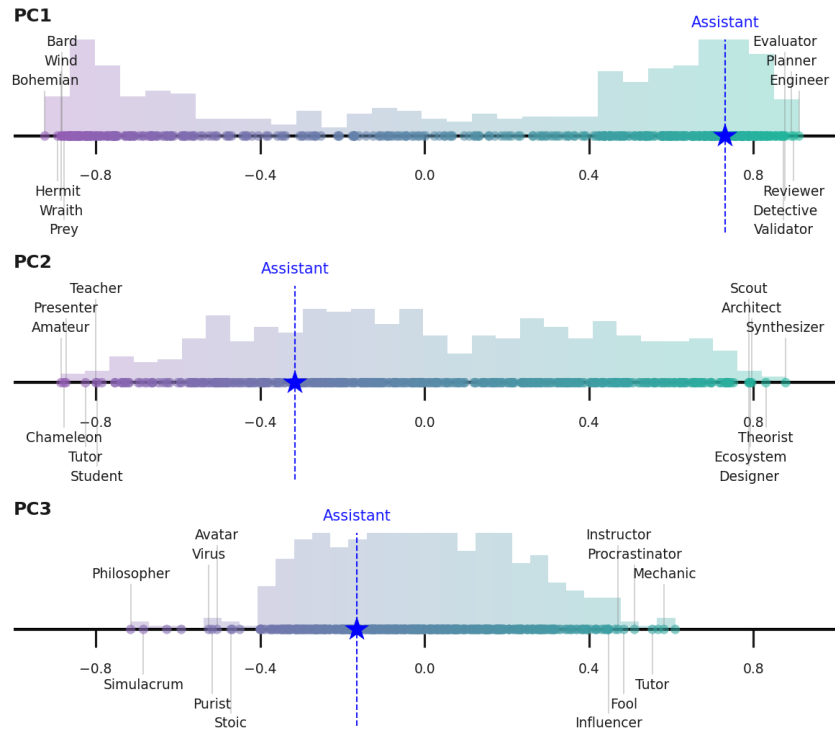


Figure 18: Histogram of cosine similarities of base Gemma 2 27B role vectors with the top 3 PCs from persona space, with selected roles (as well as the default Assistant) labeled.

C Mapping out trait space

C.1 Data generation

To map out a persona space in terms of traits rather than roles, we used a list of 240 traits that covers a wide range of possible character traits an interlocutor can take on (e.g. *speculative*, *dramatic*, *acerbic*).

We used Chen et al. [11]’s instruction generation pipeline, namely relying on a frontier model to create contrastive system prompts encouraging and discouraging each trait. For every trait, we generated five pairs of system prompts: a positive prompt designed to elicit the desired trait and a negative prompt designed to suppress it.

Then, we generated 2400 rollouts for each of the positive- and negatively-elicited traits and used a scoring system ranging from 0 (not expressed) to 100 (strongly expressed) to judge how much a response exhibits the target trait.

We kept the traits with at least ten system prompt pairs with a score difference of at least 50. Then, we collected the post-MLP residual stream activation of all response tokens and subtracted the mean negatively-elicited response activation from the mean positively-elicited activation to obtain our trait vectors. Similarly to the role space discussed in the main body of this paper, we ran PCA on these 240 trait vectors to interpret the main dimensions of variation within this "trait space."

C.2 Variance explained

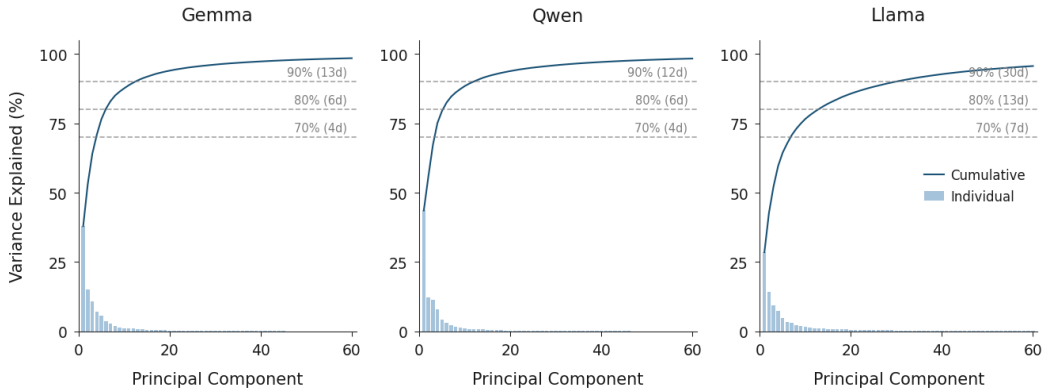


Figure 19: Variance explained by each PC for trait space. Gemma 2 27B has 239 components, Qwen 3 32B and Llama 3.3 70B have 240 components total. Trait space is also relatively low-dimensional: both Gemma and Qwen need four and Llama needs seven dimensions to explain 70% of variance.

C.3 PC interpretation

The PCs of trait space come with more similarities across models, which we again calculated by looking at the trait compositions of the top PC’s. No pairwise correlation in the top 3 PC’s was below 0.70. For the labels in the table, we opted for a cutoff of 0.85 for a shared interpretation.

Across all three models, the pairwise similarity on trait PC1 is >0.81 , reflecting an axis that roughly spans *conscientious* to *impulsive* characteristics. Llama’s PC1 is slightly distinct from the other two in that the axis is more outright *agreeable* to *antagonistic*. Gemma and Qwen’s (0.93), on the other hand, appears more calm and *regulated* to edgy and *reactive*. Like the PC1 in role space, this axis also seems to suggest traits we expect the Assistant to exhibit on one end and traits we expect to be discouraged on the other.

PC2 is moderately consistent across models (pairwise correlations >0.70), with Qwen and Llama’s (0.86) both spanning *analytical* to *intuitive* traits, though Llama also comes across as reserved to

dramatic. Gemma’s PC2 is somewhat different, spanning more *understated* to *expansive* traits: one end reflects efficiency while the other end captures curious, exploratory energy.

Qwen and Llama’s PC3 looks like an *accessible* to *erudite* axis with warm, approachable traits on one end and specialized, intellectual traits on the other. They are once again highly similar to each other (0.91), though Qwen’s erudite end feels somewhat colder (*cynical*, *callous*). Gemma’s PC3 (correlation with the other two >0.72) could also be interpreted as spanning accessible to erudite, but comes with the added connotation of *flexible* to *imposing* traits.

Table 7: The top 5 traits with the highest and lowest cosine similarity with the top three trait PCs for each model. Bolded traits are shared across all three models while italicized traits appear for two.

	Gemma 2 27B		Qwen 3 32B		Llama 3.3 70B	
	–	+	–	+	–	+
Trait PC1	REGULATED ↔ REACTIVE					
	<i>methodical</i>	<i>edgy</i>	<i>calm</i>	<i>mercurial</i>	diplomatic	flippant
	<i>calm</i>	whimsical	solemn	<i>edgy</i>	supportive	impulsive
	conscientious	<i>mercurial</i>	serious	temperamental	conscientious	visceral
	moderate	zealous	<i>methodical</i>	irreverent	strategic	petty
	structuralist	goofy	conscientious	manic	earnest	bitter
Trait PC2	UNDERSTATED ↔ EXPANSIVE			ANALYTICAL ↔ INTUITIVE		
	<i>literal</i>	charismatic	<i>literal</i>	<i>rhetorical</i>	reserved	<i>rhetorical</i>
	efficient	curious	materialist	eloquent	dispassionate	dramatic
	concise	generous	avoidant	philosophical	detached	<i>metaphorical</i>
	closure-seeking	exploratory	<i>grounded</i>	<i>metaphorical</i>	rationalist	poetic
	<i>understated</i>	proactive	understated	<i>artistic</i>	<i>grounded</i>	<i>artistic</i>
Trait PC3	FLEXIBLE ↔ IMPOSING			ACCESSIBLE ↔ ERUDITE		
	improvisational	dominant	benevolent	<i>esoteric</i>	<i>casual</i>	technical
	<i>accommodating</i>	arrogant	forgiving	cynical	chill	<i>bombastic</i>
	empathetic	<i>esoteric</i>	<i>accommodating</i>	callous	accessible	<i>specialized</i>
	<i>gregarious</i>	perfectionist	empathetic	<i>specialized</i>	empathetic	ritualistic
	casual	<i>bombastic</i>	<i>gregarious</i>	<i>erudite</i>	forgiving	<i>erudite</i>

C.4 Cosine similarity of top 3 trait PCs

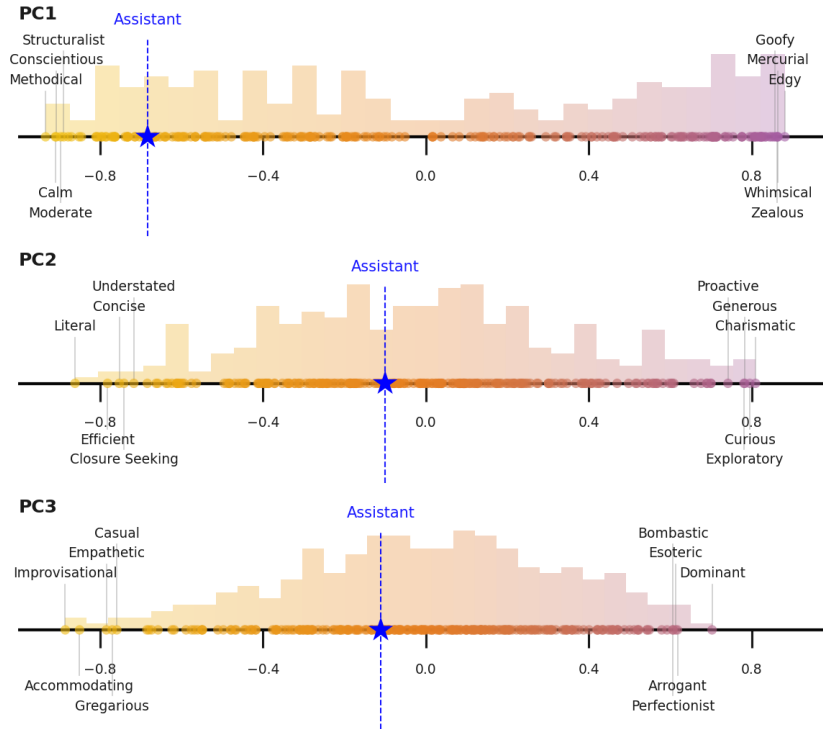


Figure 20: Histogram of cosine similarities of Gemma 2 27B trait vectors with the top 3 PCs from trait space, with selected traits.